# Experiment 6

## AIM: Set Up NGINX Load Balancing

```
Step 1: Download from https://git-scm.com

Step 2: Akshar@v19 MINGW64 ~ (main)$ node -v

v22.13.1

Step 3: Create a Project

Akshar@v19 MINGW64 /d $ mkdir load-balance-demo

Akshar@v19 MINGW64 /d $ cd load-balance-demo

Akshar@v19 MINGW64 /d/load-balance-demo $ npm init -y

Wrote to D:\load-balance-demo\package.json:


{

  "name": "load-balance-demo",

  "version": "1.0.0",

  "main": "index.js",

  "scripts": {

    "test": "echo \"Error: no test specified\" && exit 1"

  },

  "keywords": [],

  "author": "",

  "license": "ISC",

  "description": ""

}

Akshar@v19 MINGW64 /d/load-balance-demo $ npm install express

added 65 packages, and audited 66 packages in 3s

22 packages are looking for funding
```

```
  run `npm fund` for details
found 0 vulnerabilities
```

Step 4: Create server.js

Server.js for server1

```
const express = require("express");
const app = express();
const PORT = process.env.PORT || 3000;
app.get("/", (req, res) => {
res.send(`Response from Server running on port ${PORT}`);
});
app.listen(PORT, () => {
console.log(`Server running on port ${PORT}`);
});
```

Server.js for server2

```
const express = require("express");
const app = express();
const PORT = process.env.PORT || 3001;
app.get("/", (req, res) => {
res.send(`Response from Server running on port ${PORT}`);
});
app.listen(PORT, () => {
console.log(`Server running on port ${PORT}`);
});
```

Run the application

```
node server.js
```

Create a Folder Load Balancing

Add node.js project

change index.js file (server – 1)

port : 3000

Add node.js project

change index.js file (server – 2)

port : 3001

Run Multiply Servers

Terminal 1:

At PORT=3000

node server.js

Terminal 2:

at PORT=3001

node server.js

So we have now:

http://localhost:3000

http://localhost:3001

Step 5: Configure nginx for load balancing

```
http {
upstream myapp {
server localhost:3000;
server localhost:3001;
}

server {
```

```
listen 8081;
server_name localhost;
location / {
proxy_pass http://myapp;
}
}
}
```

Restart Nginx

```
nginx -s stop
Nginx
```

Now open:

http://localhost:8081

Refresh multiple times

You can check:

Response from Server running on port 3000

Response from Server running on port 3001

Make Server 1 slow

Modify Server running on PORT 3000

```
app.get("/", (req, res) => {
setTimeout(() => {
res.send("Slow Server - Port 3000");
}, 5000);
});
```

So Every request will wait 5 seconds before responding.

```
Keep Server 2 normal

app.get("/", (req, res) => {

res.send("Fast Server - Port 3001");

});


Do Testing

Open: http://localhost:8080


Now refresh quickly multiple times.


Even though server 3000 is slow,

NGINX still keeps sending every alternate request to it.


Step 5:Now Change default algo of nginx(i.e. from round robin) to Least Connections and
       test again


Modify nginx.conf file:


upstream myapp {

least_conn;

server localhost:3000;

server localhost:3001;

}

       Restart NGINX: (use below commands)

nginx -s stop

nginx


Open: http://localhost:8080

Refresh quickly multiple times quickly.
```

When slow server (3000) is busy for 5 seconds:

It has an active connection.

So NGINX sees:

Server 3000 → 1 active connection

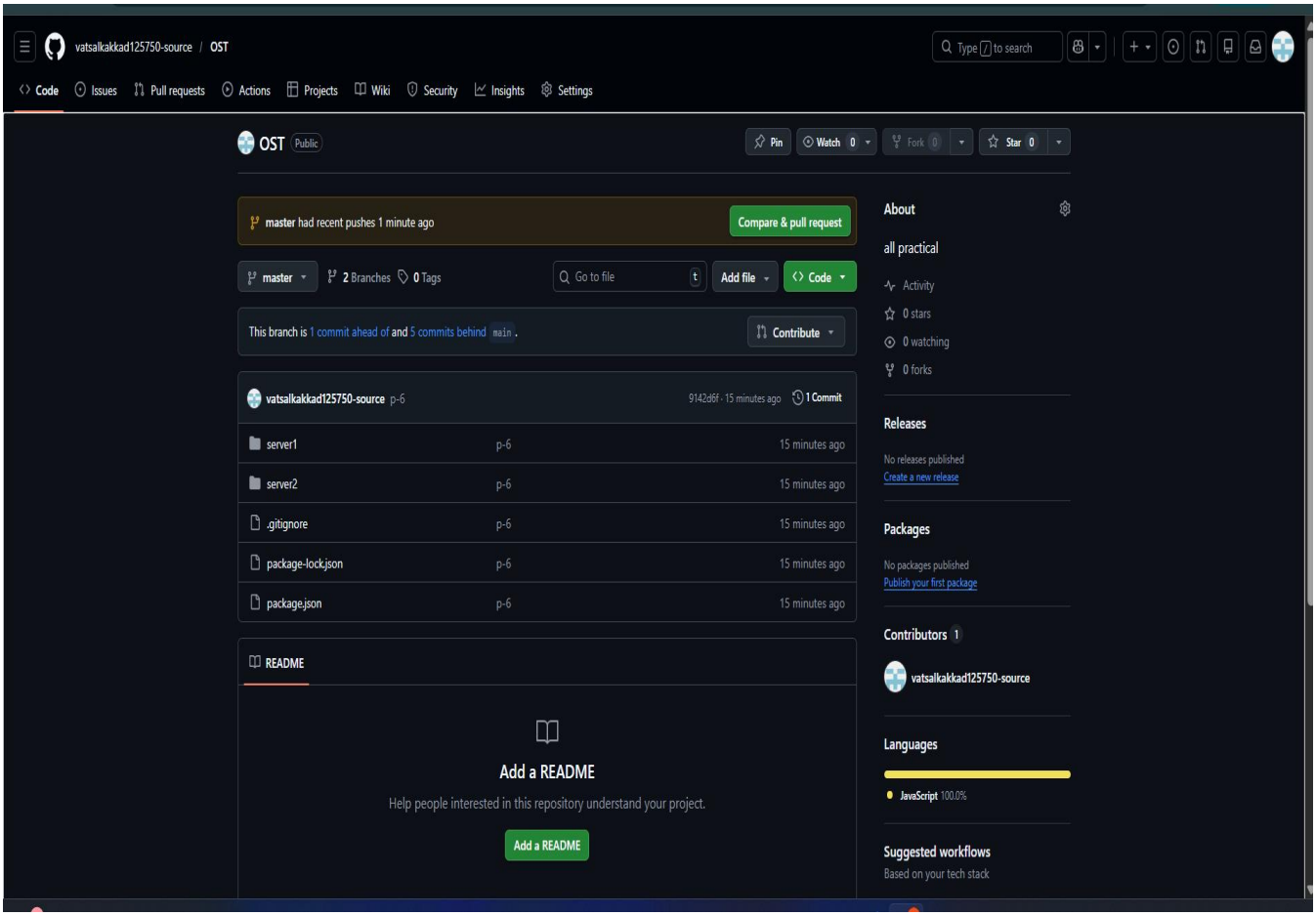
Server 3001 → 0 active connections

So it sends new requests to:

Fast Server (3001)

Output:

```
Akshar@v19 MINGW64 /d/load-balance-demo (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   package-lock.json
        new file:   package.json
        new file:   server1/package-lock.json
        new file:   server1/package.json
        new file:   server1/server.js
        new file:   server2/package-lock.json
        new file:   server2/package.json
        new file:   server2/server.js


Akshar@v19 MINGW64 /d/load-balance-demo (master)
$ git commit -m "p-6"
[master (root-commit) 9142d6f] p-6
 9 files changed, 2547 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 server1/package-lock.json
 create mode 100644 server1/package.json
 create mode 100644 server1/server.js
 create mode 100644 server2/package-lock.json
 create mode 100644 server2/package.json
 create mode 100644 server2/server.js

Akshar@v19 MINGW64 /d/load-balance-demo (master)
$ git remote add origin https://github.com/vatsalkakkad125750-source/OST.git

Akshar@v19 MINGW64 /d/load-balance-demo (master)
$ git push origin -u origin master
error: src refspec origin does not match any
error: failed to push some refs to 'https://github.com/vatsalkakkad125750-source/OST.git'

Akshar@v19 MINGW64 /d/load-balance-demo (master)
$ git push -u origin master
To https://github.com/vatsalkakkad125750-source/OST.git
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/vatsalkakkad125750-source/OST.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

Akshar@v19 MINGW64 /d/load-balance-demo (master)
$ ^C

Akshar@v19 MINGW64 /d/load-balance-demo (master)
$ git push -u origin master --force
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 8.10 KiB | 8.10 MiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/vatsalkakkad125750-source/OST.git
 + cd45a3e...9142d6f master -> master (forced update)
branch 'master' set up to track 'origin/master'.
```


vatsalkakkad125750-source / OST
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
OST
Public
Pin
Watch 0
Fork 0
Star 0
master had recent pushes 1 minute ago
Compare & pull request
master
2 Branches
0 Tags
Go to file
t
Add file
Code
This branch is 1 commit ahead of and 5 commits behind main .
Contribute
vatsalkakkad125750-source p-6
9142d6f · 15 minutes ago
1 Commit
server1 p-6 15 minutes ago
server2 p-6 15 minutes ago
.gitignore p-6 15 minutes ago
package-lock.json p-6 15 minutes ago
package.json p-6 15 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
all practical
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
vatsalkakkad125750-source
Languages
JavaScript 100.0%
Suggested workflows
Based on your tech stack